

Thermoelectric and Magnetothermoelectric Transport Measurements of Graphene

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The conductance and thermoelectric power (TEP) of graphene is simultaneously measured using a microfabricated heater and thermometer electrodes. The sign of the TEP changes across the charge neutrality point as the majority carrier density switches from electron to hole. The gate dependent conductance and TEP exhibit a quantitative agreement with the semiclassical Mott relation. In the quantum Hall regime at a high magnetic field, quantized thermopower and Nernst signals are observed and are also in agreement with the generalized Mott relation, except for strong deviations near the charge neutrality point.

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Recent research on graphene, a zero-mass, zero-gap, ambipolar material system, has focused on its unique electronic properties [1]. Although there have been theoretical investigations of thermal and thermoelectric properties of this two-dimensional material [2–8], only an indirect measurement of the thermal conductivity [9] employing Raman spectroscopy has been reported recently, and thermoelectric transport has yet to be explored experimentally. In general, the thermoelectric power (TEP) is of great interest in understanding electronic transport due to its extreme sensitivity to the particle-hole asymmetry of a system [10,11]. Measurement of the thermoelectric properties of graphene can thus elucidate details of the electronic structure of the ambipolar nature of graphene that cannot be probed by conductance measurements alone. Furthermore, employing the semiclassical Mott relation [10], one can verify the validity of the Boltzmann approach to transport in graphene by comparing the simultaneously measured conductance and TEP at different chemical potentials, similar to the experiment on carbon nanotubes [12]. The application of a magnetic field in addition to a thermal gradient provides a valuable experimental tool to investigate magnetothermoelectric effects, such as the Nernst effect in two-dimensional electronic systems in the quantum Hall (QH) regime [13–16].

In this Letter, we first discuss the temperature and carrier density dependence of the TEP measured in mesoscopic graphene samples, and its relation with the Mott formula. We then discuss the magnetothermopower measurement where we observe unusual quantized TEP in the presence of the QH effect.

Graphene samples are fabricated using the mechanical exfoliation method on 300 nm SiO₂/Si substrate. The graphene carrier density is adjusted with applied gate voltage to the degenerately doped Si substrate. Electrodes are defined with standard electron beam lithography, followed by Ti/Au (3/30 nm) evaporation and liftoff processes. The TEP measurement technique was described in

detail elsewhere [12]. In brief, a controlled temperature gradient ΔT is applied to the sample by a microfabricated heater while the resulting thermally induced voltage ΔV is measured by the voltage probes to acquire the TEP, $S = -\frac{\Delta V}{\Delta T}$ (see the upper inset of Fig. 1 for device layout). Local temperature variations are measured with two metal four-probe microthermometers. Oxygen plasma etching is used to isolate the graphene flake electrically from the heater electrode. We use Raman spectroscopy and QH measure-

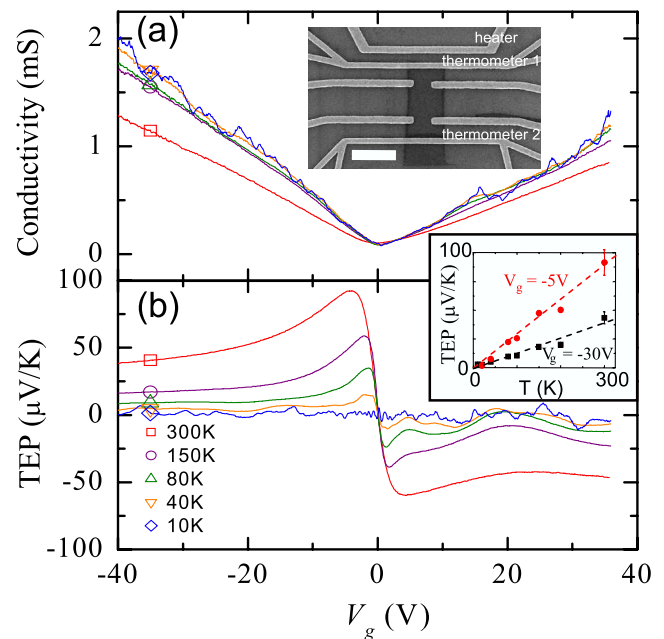


FIG. 1 (color online). (a) Conductivity and (b) TEP of a graphene sample as a function of V_g for $T = 300$ K (square), 150 K (circle), 80 K (up triangle), 40 K (down triangle), and 10 K (diamond). Upper inset: Scanning electron microscopy image of a typical device, the scale bar is $2 \mu\text{m}$. Lower inset: TEP values taken at $V_g = -30$ V (square) and -5 V (circle). Dashed lines are linear fits to the data.

石墨烯的热电和磁热电输运测量

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通过微加工加热器与温度计电极同步测量石墨烯的电导率和热电势 (TEP)。当多数载流子密度从电子转变为空穴时, TEP的符号会在电荷中性点处发生改变。栅极依赖的电导率与TEP与半经典莫特关系呈现定量一致性。在强磁场的量子霍尔效应区域, 观测到量子化热电势和能斯特信号, 这些现象同样符合广义莫特关系, 仅在电荷中性点附近存在显著偏差。

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近期关于石墨烯 (一种零质量、零能隙、双极性材料体系) 的研究主要聚焦于其独特的电子特性[1]。尽管已有理论研究探讨了这种二维材料的热学与热电特性[2-8], 但近期仅报道了采用拉曼光谱法间接测量热导率[9]的结果, 而热电输运尚未通过实验进行探索。总体而言, 热电势 (TEP) 因其对系统中粒子-空穴不对称性的极端敏感性, 在理解电子输运方面具有重要价值[10, 11]。因此, 通过测量石墨烯的热电特性, 可以揭示其双极性本质的电子结构细节——这些细节仅凭电导测量无法探测。此外, 通过运用半经典莫特关系式[10], 可以验证玻尔兹曼方法在石墨烯输运中的有效性——通过比较不同化学势下同时测量的电导率和热电势, 这与碳纳米管实验[12]类似。在热梯度基础上叠加磁场的应用, 为研究磁热电效应 (如量子霍尔 (QH) 区域二维电子系统中的能斯特效应) 提供了有价值的实验工具[13-16]。

本文首先讨论了在介观石墨烯样品中测得的热电势 (TEP) 随温度和载流子密度的变化规律, 及其与莫特公式的关系。随后, 我们探讨了磁热电势的测量结果, 在量子霍尔效应存在的情况下, 观察到了异常的量子化热电势现象。

石墨烯样品采用机械剥离法在300纳米SiO₂=Si基底上制备。通过施加栅极电压对简并掺杂的Si基底进行石墨烯载流子密度调节。电极采用标准电子束光刻技术定义, 随后进行Ti=Au (3=30纳米) 蒸发和剥离工艺。TEP测量技术已在文献中描述。

详见其他部分[12]。简言之, 通过微加工加热器对样品施加受控温度梯度 ΔT , 同时测量由此产生的热致电压 ΔV 由电压探针获取以获取 TEP, $S = -\frac{\Delta V}{\Delta T}$ (器件布局参见图1上部插图)。采用两根金属四探针微量热计测量局部温度变化。通过氧等离子体刻蚀技术实现石墨烯薄片与加热电极的电隔离。我们运用拉曼光谱技术及量子霍尔效应进行测量。

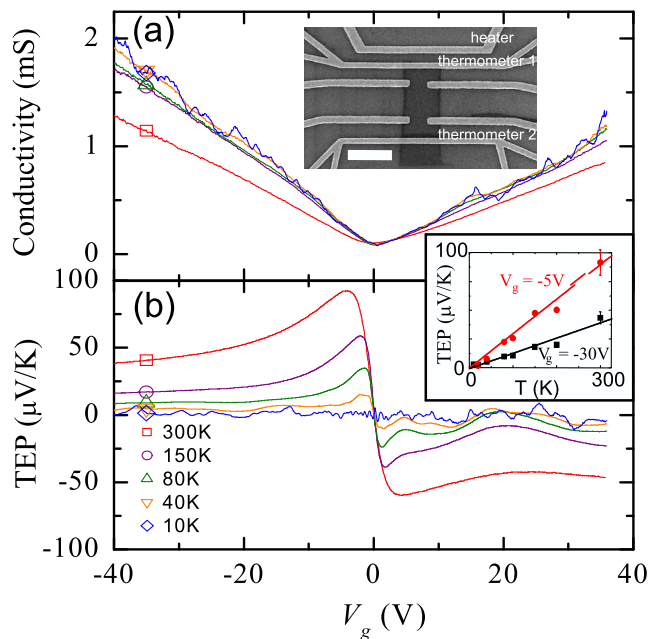


图1 (彩色在线)。(a)电导率与(b)热电势 (TEP) 的曲线, 对应于 graphene 150 K (square), 100 K (circle), 80 K (up triangle), 40 K (down triangle), 10 K (diamond)。上插图: 典型器件的扫描电子显微镜图像, 比例尺为 2 μm 。下插图: TEP values taken at $V_g = -30\text{ V}$ (circle). Dashed lines are linear fits to the data (square) and (diamond) and (circle).

ments in order to verify that our samples are single layer graphene [17,18]. Devices are annealed in a vacuum cryostat to 400 K in order to improve mobility and electron-hole symmetry. We measure both the two-terminal mesoscopic conductance G utilizing the outer electrodes and the four-terminal conductivity σ employing the Hall bar geometry. To measure the TEP we apply a low frequency ac signal to the heater electrode and detect the resulting 2ω thermovoltage from the outer electrodes as we change the graphene carrier density with the backgate. To stay in the linear response regime the condition $\Delta T \ll T$ is always satisfied.

In this work, we study 10 different graphene samples with mobility ranging from 1000 to 7000 cm²/Vs. Figures 1(a) and 1(b) show simultaneously measured σ and TEP as a function of applied gate voltage V_g for a representative graphene device in the temperature range of 10–300 K. As it was shown previously [1,19], $\sigma \propto V_g$ away from the charge neutrality point (CNP). The conductivity becomes minimum at the CNP, corresponding to $V_g = V_D$, where $V_D \approx 0$ V for this particular device. The peak value of TEP reaches ~ 80 μ V/K at room temperature in the majority of devices measured. Notably, the sign of the TEP, which indicates the sign of the majority charge carrier, changes from positive to negative as V_g crosses the CNP. The electron-hole asymmetry observed in the conductance and also in the TEP is most likely due to charged impurities in the oxide substrate [19,20] or the formation of a p - n junction at the electrode contacts [21]. Universal conductance fluctuations are observed in the conductance for $T \lesssim 30$ K and are quantitatively correlated to the oscillations in the TEP, as we will show later.

While σ decreases as T increases, the magnitude of the TEP increases monotonically as T increases. The inset of Fig. 1(b) shows a linear temperature dependence of TEP at two representative gate voltages, $V_g = -30$ V and $V_g = -5$ V. The linearity of the TEP in T suggests that the mechanism for thermoelectric generation is diffusive thermopower. Phonon-drag components of the thermopower, which usually appear in the form of a nonmonotonic temperature dependence of TEP [22], are not present here, which is consistent with weak electron-phonon coupling in graphene [23,24].

Within the Boltzmann formulation, the Seebeck coefficient S , equivalent to the TEP, represents the entropy transported per unit charge. This thermodynamic relation provides us with a qualitative understanding of the overall behavior of the observed $S(V_g)$ as V_g tunes the carrier density n and thus the Fermi energy E_F of the graphene samples. Far away from the CNP, the electronic system is highly degenerate and the entropy transported in the channel is proportional to the number of thermally activated carriers over the degenerate Fermi sea, leading to $S \sim T/E_F$. Close to the charge neutrality point, however, the electron gas becomes nondegenerate and both electrons and holes are thermally populated. Here the TEP gains

opposite contributions from electrons and holes, so $|S|$ decreases as $|n|$ decreases. However, the nondegenerate limit is difficult to achieve in most of the experimental temperature range since the electron-hole puddle formation near the charge neutrality point yields a finite local density $>10^{11}$ cm⁻² [25], corresponding to ≥ 100 K of energy broadening.

For a quantitative understanding of the TEP data, we now employ the semiclassical Mott relation [10,11]. In mesoscopic graphene samples, this relation can be tested by tuning E_F using V_g :

$$S = -\frac{\pi^2 k_B^2 T}{3|e|} \frac{1}{G} \frac{dG}{dV_g} \frac{dV_g}{dE} \Big|_{E=E_F}, \quad (1)$$

where e is the electron charge and k_B is the Boltzmann constant. This formula can be evaluated by numerically differentiating the measured G with respect to V_g and considering $\frac{dV_g}{dE_F}(V_g) = \frac{\sqrt{|e|}}{C_g \pi} \frac{2}{\hbar v_F} \sqrt{|\Delta V_g|}$, where $\Delta V_g = V_g - V_D$, $\hbar$ is Planck's constant, the Fermi velocity $v_F = 10^6$ m/s, and the gate capacitance $C_g = 115$ aF/ μ m² in our device geometry [18]. A comparison between the measured TEP and that predicted by the Mott formula is plotted in Fig. 2 for temperatures $T = 15$, 40, and 200 K, where we find excellent agreement at lower temperatures. We note that Eq. (1) is in the form of the mesoscopic formula [26] where the conductance includes the contact resistance [27]. At higher temperatures, local thermal equi-

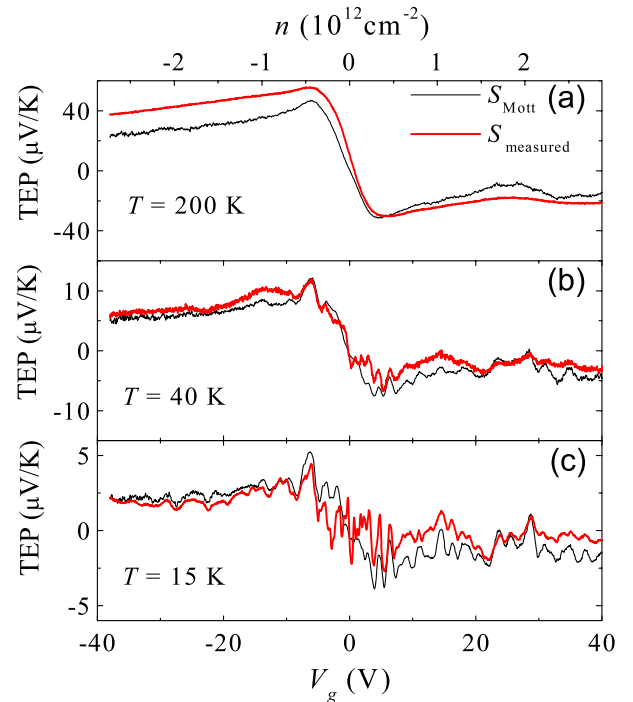


FIG. 2 (color online). Measured TEP values (thick red lines) and TEP values calculated with the Mott formula (thin black lines) using Eq. (1) as a function of applied gate voltage for $T = 200$ K (a), 40 K (b), and 15 K (c).

为验证样品是否为单层石墨烯[17, 18], 我们采用真空低温恒温器将器件退火至400K以提升载流子迁移率和电子空穴对称性。通过外电极测量双端电阻 G , 并利用霍尔条形结构测量四端电导 σ 。为测定热电势 (TEP), 我们在加热电极施加低频交流信号, 通过背栅调控石墨烯载流子密度时, 检测外电极产生的2! 热电压。为保持线性响应范围, 始终满足条件 $\Delta T \ll T$ 。

在本研究中, 我们考察了10种不同石墨烯样品, 其迁移率范围为1000至7000 cm^2/Vs 。图1(a)和1(b)同时展示了 σ 和TEP随栅极电压 V_g 变化的函数关系。在10–300 K温度范围内具有代表性的石墨烯器件。如先前所示[1, 19], $\sigma \propto V_g$ 偏离电荷中性点 (CNP)。电导率在 CNP 处达到最小值, 对应 $V_g = V_D$, 其中 $V_D \approx 0$ V (针对该特定器件)。在大多数测量的器件中, TEP 峰值在室温下达到约80 $\mu\text{V}/\text{K}$ 。值得注意的是, TEP的符号 (指示多数载流子的符号) 在 V_g 跨越时从正变负 CNP。在电导和TEP中观察到的电子-空穴不对称性最有可能是由于氧化物衬底中的带电杂质[19, 20]或电极接触处p-n结的形成[21]。在 $T \leq 30\text{K}$ 时, 电导中观察到普遍的电导波动, 如我们稍后将展示的, 这些波动与TEP中的振荡定量相关。

虽然 σ 随 T 增加而减小, 但TEP的幅值随 T 增加而单调增大。图1(b)的插图显示了 T 与 V_g 的线性依赖关系。gate voltages V_g in the TEP linearity of the TEP dependence on T suggests that the TEP is dominated by the phonon-drag mechanism. The phonon-drag mechanism typically manifests as a non-monotonic temperature dependence [22], but this is not observed here, which is consistent with the weak electron-phonon coupling in graphene [23, 24].

在玻尔兹曼公式中, 塞贝克系数 S (等效于热电子产额TEP) 表示单位电荷所携带的熵。这一热力学关系为我们提供了定性理解: 当 V_g 调节载流子密度 n (进而改变石墨烯样品的费米能 E_F) 时, 观测到的 $S(V_g)$ 整体行为。远离 CNP 时, 电子系统高度简并, 通道中传输的熵与简并费米海中热激活载流子的数量成正比, 因此 $S \propto T/E_F$ 。然而在接近电荷中性点时, 电子气变得非简并, 电子和空穴均被热激发占据。此时热电子产额TEP获得

电子与空穴的贡献相反, 因此 S 随 n 减小而减小。然而, 在大多数实验温度范围内难以实现非简并极限, 因为电荷中性点附近的电子-空穴池形成会产生有限的局域密度 $>10^{11} \text{cm}^{-2}$ [25], 这对应于 $\geq 100 \text{K}$ 的能量展宽。

为定量理解TEP数据, 我们现采用半经典Mott关系[10, 11]。在介观石墨烯样品中, 可通过调节 V_g 来调整 E_F 以验证该关系:

$$S = - \frac{\pi^2 k_B^2 T}{3|e|} \frac{1}{G} \frac{dG}{dE} \bigg|_{E=E_F} \quad (1)$$

其中 e 为电子电荷, k_B 为玻尔兹曼常数。该公式可通过对测量的 G 值关于 V 进行数值微分来计算。

鉴于 $\frac{dV_g}{dE_F} = \frac{\sqrt{q}}{C_g \pi \hbar v_F} \sqrt{|\Delta V_g|}$, 其中 $\Delta V_g = V_g - V_D$, $v_F = 10^6 \text{m/s}$, $C_g = 1.15 \text{aF}/\mu\text{m}^2$ in the experiment, the Fermi velocity $v_F = 10^6 \text{m/s}$, and the capacitance $C_g = 1.15 \text{aF}/\mu\text{m}^2$ in the experiment. A comparison between the measured TEP and Mott formula prediction is shown in Fig. 2, where we find excellent agreement at low temperatures. We note that the form of the Mott formula [26], which includes contact resistance [27], is valid at high temperatures, where the thermal energy

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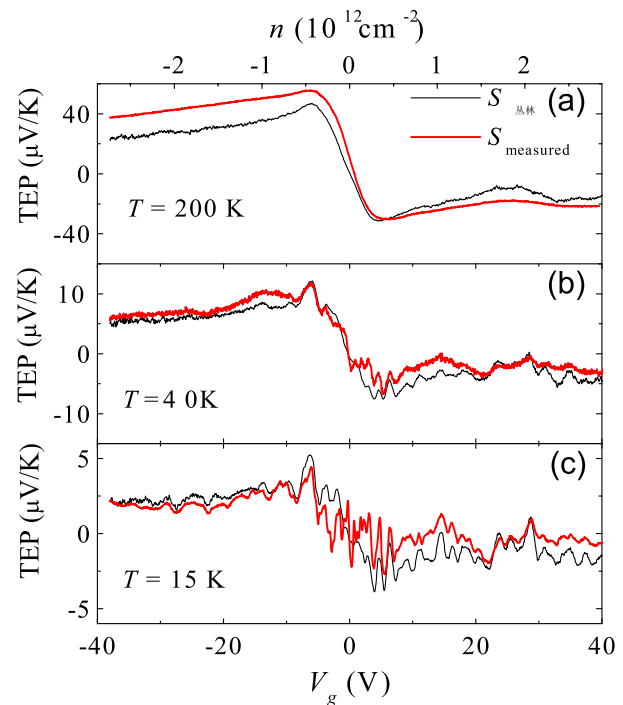


图2 (彩色在线)。使用等式 (1) 计算的TEP值 (粗红线) 与Mott公式计算的TEP值 (细黑线) 随施加栅极电压的变化关系, 对应温度 $T = 200 \text{K}$ (a)、 40K (b)和 15K (c)。

librium is established, and the mesoscopic form Eq. (1) using G becomes less accurate, as it is already seen in the 200 K data in Fig. 2. Such deviation grows at higher temperatures. Recent theoretical work [7] predicts that in graphene the enhanced inelastic scattering time of carriers in the presence of electron-electron interactions yields TEP that deviates strongly from the Mott relation. Further experimental work with higher mobility samples (in current samples disorder dominates transport even at room temperature) is needed to elucidate this nondegenerate, high temperature limit of TEP in graphene.

We now turn our attention to the magneto-TEP. In the presence of a perpendicular magnetic field, a Lorentz force bends the trajectories of the thermally diffusing carriers to produce the Nernst TEP, i.e., transverse components of the TEP, S_{yx} . For this measurement, we utilize a multiple probe configuration to measure the tensor components of resistivity R_{xx} and R_{xy} and also TEP components S_{xx} and S_{yx} as shown in the inset of Figs. 3(b) and 3(c). Figure 3(a) displays the Hall conductivity $\sigma_{xy} = -R_{xy}/[R_{xy}^2 + (WR_{xx}/L)^2]$ as a function of magnetic field and gate voltage, where W and L are the width and length of the Hall bar device, respectively. The linear growth of the oscillations of $\sigma_{xy}(V_g)$ with increasing magnetic field clearly indicates

that Shubnikov-de Hass oscillations are present in graphene as previously observed [17,18]. Remarkably, similar oscillatory features develop in both S_{xx} and S_{yx} as shown in Figs. 3(b) and 3(c).

In the high field regime, $B \sim 9$ T, fully developed QH effect is observed for filling factors $\nu = \pm 2, \pm 6, \pm 10, \pm 14$, indicated by the quantized plateaus in R_{xy} and zeros in R_{xx} . Vertical dashed lines in Figs. 3(d)–3(g) indicate boundaries between Landau levels (LLs) where we find S_{xx} and S_{yx} tend to zero since no carriers are available to participate in diffusion. At the Landau level centers $|S_{xx}|$ is maximal and as higher LL are occupied these peaks decrease. Theoretically, this quantized behavior of S_{xx} is expected in the QH regime of 2D electron gas systems [13,16]. In the low temperature limit, the peaks of S_{xx} are predicted to be given by $S_{xx}^{\max} = -\frac{k_B \ln 2}{e} \frac{1}{N}$ for clean samples [28], where N in graphene indicates the LL index. Experimentally, thermal and disorder induced broadening of LLs tends to broaden S_{xx} , making the quantized values smaller than the above prediction [15,16]. The observed $S_{xx}^{\max}$ shows the quantized $1/N$ trend discussed above, but with a reduced factor, indicating the presence of disorder in our graphene samples.

Further analysis which includes the Hall conductivity tensor and the Nernst TEP is performed by employing the generalized Mott formula that holds for all temperatures and can be applied to the QH regime [13,14]:

$$S_{ij}^{\text{Mott}} = -\frac{\pi^2 k_B^2 T}{3|e|} \sum_k (\sigma^{-1})_{ik} \left(\frac{\partial \sigma}{\partial E_F} \right)_{kj}, \quad (2)$$

where σ_{ij} is the conductivity tensor and i, j represent the x and y components. According to Eq. (2), we expect that S_{xx}/T becomes temperature independent in the QH limit where σ_{ij} is temperature independent. Indeed, Fig. 4(a) shows that the measured S_{xx}/T is rather temperature independent for $T < 20$ K. Above this temperature, however, the oscillation amplitude of S_{xx}/T becomes smaller since thermal activation across the disorder broadened LLs becomes appreciable and the QH effects disappear. Figure 4(b) shows good agreement between the measured $S_{xx}(S_{yx})$ and the calculated $S_{xx}^{\text{Mott}}(S_{yx}^{\text{Mott}})$, indicating that the semiclassical Mott relation, extended into the QH regime, works well for both holes ($N < 0$) and electrons ($N > 0$). Near the zeroth LL ($N = 0$) at the CNP ($V_g \approx V_D$), however, the measured S_{xx} exhibits a pair of anomalous oscillations that shows a distinct deviation from Eq. (2). We remark that this pair of S_{xx} peaks near the zeroth Landau level shows opposite TEP polarity, i.e., positive peak on the electron side ($V_g > V_D$) and negative peak on the hole side ($V_g < V_D$). This pair of opposite polarity oscillations could be indicative of a peculiar QH state stemming from the $N = 0$ LL. It has been speculated that this LL produces a pair of counterpropagating edge states when the Fermi level is at the CNP [29]. In this scenario, the polarity of the inner edge state is opposite to the sign of the bulk

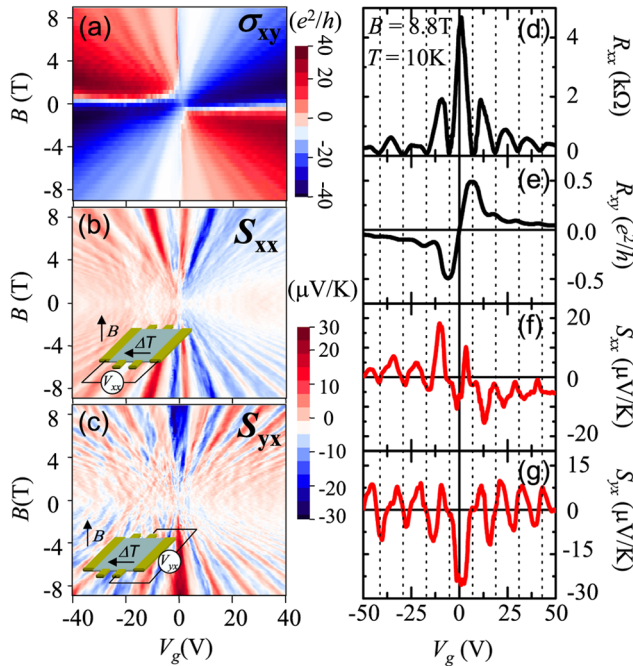


FIG. 3 (color online). (a) Hall conductance, (b) longitudinal TEP S_{xx} , and (c) transverse TEP S_{yx} as a function of V_g and B at $T = 10$ K. The insets show the electrode configurations for S_{xx} and S_{yx} measurement, respectively. (d) R_{xx} , (e) R_{xy} , (f) S_{xx} , and (g) S_{yx} as a function of V_g at a fixed magnetic field $B = 8.8$ T and $T = 4.2$ K. The vertical dotted lines indicate filling factors corresponding to $\nu = \pm 2, \pm 6, \pm 10, \pm 14$. All the data are averaged with respect to positive and negative values of magnetic field in order to remove mixing between longitudinal and transverse components.

librium已确立, 而使用G的介观形式等式(1)的准确性会降低, 这在图2的200 K数据中已经有所体现。这种偏差在更高温度下会进一步增大。近期理论研究[7]预测, 在石墨烯中, 电子-电子相互作用下载流子的增强非弹性散射时间会导致TEP与莫特关系产生显著偏差。需要进一步使用更高迁移率样品的实验研究(在当前样品中, 无序性甚至在室温下也主导了输运)来阐明石墨烯中TEP的这种非简并高温极限。

我们现在将注意力转向磁热电势。在垂直磁场存在的情况下, 洛伦兹力会使热扩散载流子的轨迹发生偏转, 从而产生能斯特热电势, 即热电势的横向分量 S_{yx} 。为此, 我们采用多探针配置来测量电阻率的张量分量 R_{xx} 和 R_{xy} , 以及热电势分量 S_{xx} 和 S_{yx} , 如图3(b)和3(c)插图所示。图3(a) (是 σ_{xy}/L^2) as a function of activity $\sigma_{xy} = -R_{xy}/[R_{xx}^2 + R_{xy}^2]$, 其中W和L分别表示霍尔条器件的宽度和长度。 σ_{xy} (Vg) 振荡随磁场增强的线性增长清晰表明

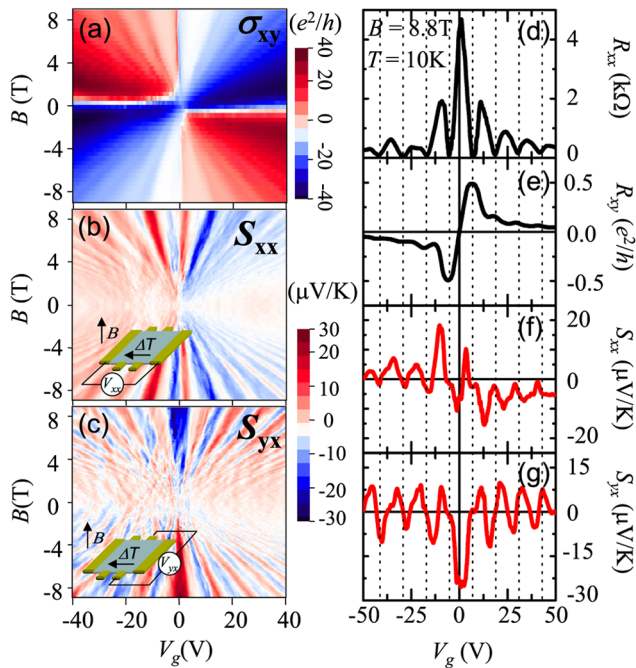


图3 (彩色在线)。(a)霍尔电导, (b)纵向TEP S_{xx} , 以及(c)横向TEP S_{yx} 随 V_g 和 B 在

T 处的变化 $\frac{1}{S} \frac{dS}{dT} = \frac{1}{S} \frac{dS}{dV_g} \frac{dV_g}{dT} + \frac{1}{S} \frac{dS}{dB} \frac{dB}{dT}$ Tashuer i nsetment, show the elerespectively c t (rd o) de c x, o n (e fi) guRr x a, ti (of n) s fo xx r, Sxanxd ac gno) drrSyTes = 4.2 pondin function f V_g at a fi xed magne eld = K The 卵ertical dotted lines ind 词 fillTBinag fe8aca 8toveTrsraged 针对磁场正 g to $\nu = \pm 2, \pm 6, \pm 10, \pm 14$ Al da ar 负值以消除纵向与横向分量间的混合。

如先前观察到的那样, 石墨烯中存在Shubnikov-de Hass振荡[17, 18]。值得注意的是, 如图3(b)和3(c)所示, S_{xx} 和 S_{yx} 中均出现了类似的振荡特征。

在高场区域 ($B \sim 9$ T), 当填充因子 $\nu = \pm 2; \pm 6; \pm 10; \pm 14$ 时, 可观察到完全发展的量子霍尔效应, 这由 R_{xy} 中的量子化平台和 R_{xx} 中的零值所指示。图3(d)–3(g)中的垂直虚线标示了朗道能级 (LLs) 之间的边界, 在此边界处我们发现 S_{xx} 和 S_{yx} 趋于零, 因为没有可用的载流子。

参与扩散。在朗道能级中心 $|S_{xx}|$ 达到最大值, 随着更高能级被占据, 这些峰逐渐减小。理论上, S_{xx} 的这种量子化行为在二维电子气系统的量子霍尔效应区域中是预期的[13, 16]。

在低温极限下, S_{xx} 的峰为 $\frac{\pi^2 k_B^2 T}{3|e|} \sum_k (\sigma^{-1})_{ik} \left(\frac{\partial \sigma}{\partial E_F} \right)_{kj}$, where N in graph $\sigma_{xx}^{\max} = -\frac{k_B}{e} \frac{n_2}{N}$ fo le sa s ene in dic拉 tes rthce anLL mpleindex 实验表明, 热

致和无序诱导的LLs展览会拓宽 S_{xx} , 使得量子化值小于上述

预测[15, 16]。观测到的 S_{xx} 显示了上述讨论的 $1/N$ 量子化趋势, 但因子减小, 表明我们的石墨烯样品中存在无序。

采用适用于所有温度且可应用于量子霍尔效应区的广义莫特公式[13, 14], 进行了包括霍尔电导张量和能斯特热电势在内的进一步分析:

$$S_{ij}^{\text{Mott}} = -\frac{\pi^2 k_B^2 T}{3|e|} \sum_k (\sigma^{-1})_{ik} \left(\frac{\partial \sigma}{\partial E_F} \right)_{kj}, \quad (2)$$

其中 σ_{ij} 是电导率张量, i, j 分别代表 x 和 y 分量。根据等式

(2), 我们预期在 σ_{ij} 与温度无关的量子霍尔极限下, $S_{xx} = T$ 将变得与温度无关。事实上, 图4(a)显示, 当温度 $T < 20$ K时, 测量得到的 $S_{xx} = T$ 表现出较强的温度无关性。然而当温度超过此值时, $S_{xx} = T$ 的振荡幅度会减小, 因为无序能带间的热激活效应变得显著, 量子霍尔效应随之消失。图4(b)展示了测量得到的 S_{xx} (S_{yx}) 与计算得到的 S_{xx}^{Mott} (S_{yx}^{Mott}) 之间具有良好的一致性, 表明扩展到量子霍尔区域的半经典莫特关系对空穴 ($N < 0$) 和电子 ($N > 0$) 均适用。Nevertheless, the zeroth LL ($N_{xx} = 0$) 抑制钙离子通道 (Ca²⁺通道) 的开放 $\approx \frac{1}{m} \text{alVD} \text{ous} \text{'hoows cil} \text{'-ations}$ 显示与等式(2)的明显偏离。我们注意到这对位于零朗道能级附近的 S_{xx} 峰表现出相反的TEP极性, 即电子侧为正峰 ($V_g > V_D$), 空穴侧为负峰 ($V_g < V_D$)。这种相反极性的振荡可能表明一种源自Npa的特殊量子霍尔态。 $= 0$ LLco. It haerp s been speculat ir of LLco unt haerp s propagating ed ed that ts hi s LL pro当 duFeersmai水平处于 CNP [29]时。 ge state 在此场景中, 内边缘态的极性与体材料符号相反

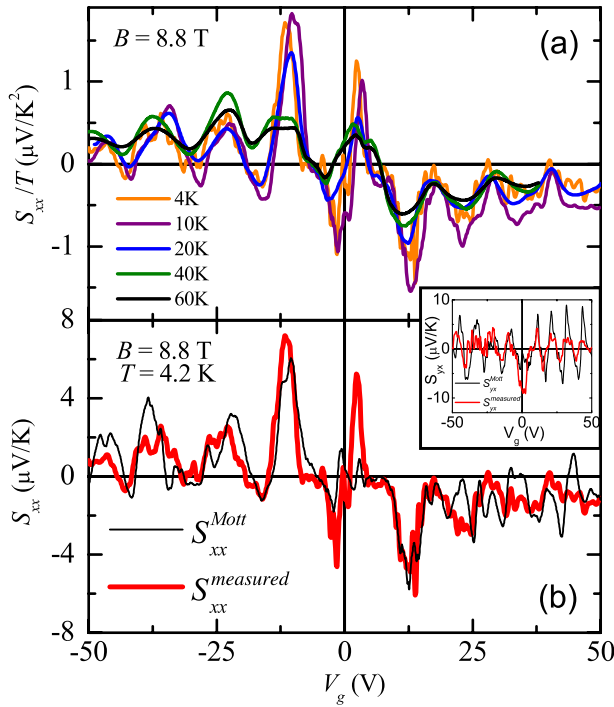


FIG. 4 (color online). (a) Temperature dependence of the S_{xx}/T as a function of V_g at $B = 8.8$ T. (b) Measured S_{xx} (thick red line) and the S_{xx} predicted by the generalized Mott relation (thin black line) at $B = 8.8$ T and $T = 4.2$ K showing deviations near the $N = 0$ LL. Inset: Corresponding S_{yx}^{measured} and S_{yx}^{Mott} .

majority carrier, which may generate the anomalous TEP observed. We also note that an alternative explanation of this feature is possible based on the sample shape dependent two-terminal QH conductance variations near the CNP [30]. In addition, a large enhancement of S_{yx} [inset of Fig. 4(b)] is observed near the CNP, which also deviates strongly from S_{yx}^{Mott} . Such enhancement of the Nernst signal is predicted in conventional 2D electron systems, where the magnitude of S_{yx} strongly depends on the disorder strength [13,14].

In conclusion, we report carrier density dependent measurements of the thermoelectric coefficient in graphene which scales linearly with temperature. The semiclassical Mott relation explains the observed carrier density modulated TEP. In the QH regime at high magnetic field, the tensor components of the TEP show quantized behavior. Because of its sensitivity to electron-hole asymmetry, TEP measurements provide a complementary experimental tool to further study quantum transport phenomena in high mobility graphene samples.

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Note added.—During the preparation of this manuscript, we became aware of a related work with similar experimental results [31].

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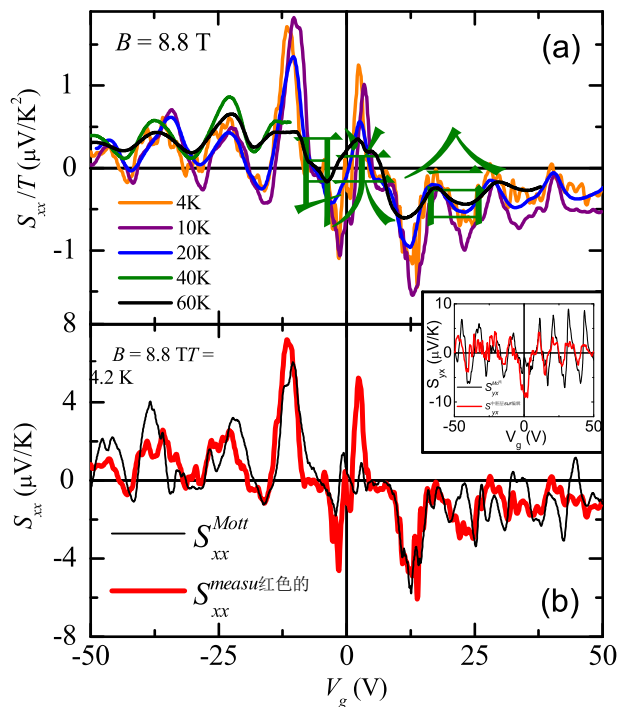


图4 (彩色在线)。(a)在 $B = 8.8 \text{ T}$ 时, S_{xx}/T 随 V_g 变化的温度依赖性。(b)实测 S_{xx} (粗红线)与广义莫特关系 $S_{xx}^{\text{Mott}} = \frac{1}{e} \frac{dS_{yx}}{dV_g}$ (细黑线)在 $B = 8.8 \text{ T}$ 和 $T = 4.2 \text{ K}$ 时的对比。Inset: dS_{xx}/dV_g 的对应关系。

多数载流子,这可能是观察到的异常TEP产生的原因。我们还注意到,基于样品形状依赖的双端量子霍尔电导在CNP附近的变异[30],这一特征可能存在另一种解释。此外,在CNP附近观察到 S_{yx} 的显著增强[图4(b)插图],这也与 S_{xx}^{Mott} 有显著偏离。这种能斯特信号的增强在传统二维电子系统中可预测,其中 S 的大小强烈依赖于无序强度[13,14]。

综上所述,我们通过载流子密度依赖性实验,首次测得石墨烯热电系数随温度呈线性变化的特性。采用半经典Mott关系式,成功解释了载流子密度调控下的热电系数变化规律。在强磁场量子霍尔效应区域,热电系数的张量分量呈现出量子化行为。由于对电子-空穴不对称性具有高度敏感性,该热电系数测量方法为深入研究高迁移率石墨烯样品中的量子输运现象提供了重要实验手段。

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注释补充——在撰写本手稿期间,我们注意到一项具有相似实验结果的相关研究[31]。

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